Breederify: Using Siamese Neural Networks for Similarity-Based Dog Breed Classification

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***Abstract*—**Machine learning systems find it difficult to differentiate between dog breeds because dog breeds exhibit physical resemblances to their counterparts and dog breeds display different physical characteristics. The typical Convolutional Neural Network (CNN) system requires extensive image dataset processing and complete system redevelopment and high-performance server GPU hardware which makes it unsuitable for use in small-scale environments such as animal shelters and veterinary clinics.

The project presents "Breederify" which operates as a Lightweight Siamese-CNN system that combines Metric Learning with a MobileNetV2 backbone to enable One-Shot Learning for dog breed identification. The system enables the network to learn how dogs share similar features which researchers evaluate through Euclidean distance measurements because the system requires only one breed reference image for identification. The machine learning core function uses a restricted "micro-dataset" which contains between 5 and 10 images from both public domains and smartphones for each breed. The training process included data augmentation together with L2 Normalization for feature embedding stabilization and Contrastive Loss and backbone layer top tuning to achieve precise capture of dog breed characteristics.

The model functions on standard consumer hardware because its design accommodates systems with 4GB VRAM limits while delivering practical reliability through efficient operation. The system assessed its reliability through a calibrated similarity threshold test which achieved True Accuracy at 92.50% and Equal Error Rate (EER) at 5.00%. The automated solution which operates at low cost demonstrates how Siamese networks function effectively in environments where data availability is extremely limited, which supports animal welfare organizations.

Keywords— Siamese Neural Networks, Metric Learning, Dog Breed Classification, MobileNetV2, Animal Welfare Technology.

I. INTRODUCTION

Identifying dog breeds presents a classic challenge in the field of machine learning because dogs of the same breed show more physical differences than between two different breeds. The traditional convolutional neural network (CNN) models need to process thousands of training images for each breed because they require large amounts of data to achieve trustworthy results. The standard CNNs need complete system upgrades through major dataset additions when they want to include new dog breeds into their framework. The training of these conventional models depends on powerful server-grade GPU hardware.

Researchers need essential components to build their project because they need project materials for construction. Multiple dog breeds need identification through these rapid and inexpensive methods which animal welfare organizations and local shelters and veterinary clinics need to use for their administrative operations and dog adoption procedures and medical recordkeeping. The need for specialized equipment together with extensive data requirements creates major challenges which prevent small organizations and students and independent researchers from accessing research materials.

The research establishes Breederify as a solution to these limitations through its development of a Lightweight Siamese-CNN system which operates on common consumer hardware. The study employs Metric Learning through a Siamese "twin" architecture to train the network for identifying "similarity" among canine features instead of using fixed categorical labels. The system achieves One-Shot Learning which enables breed identification from a single reference image. The proposed model uses MobileNetV2 as its feature extraction backbone to identify specific dog breeds from a manually compiled dataset which contains between 5 and 10 images for each breed. The research demonstrates that automated breed identification reaches high accuracy levels in situations where data and computing power are restricted and the system does not need any specialized equipment.

II. RELATED WORKS

According to recent studies on animal welfare [1], local animal shelters and veterinary clinics frequently require rapid and accurate breed identification to assist with administrative logging and adoption processes. Out of millions of dogs entering shelters annually, accurately classifying their breeds is a classic challenge due to high intra-class variation and low inter-class variation [2]. Because of this, many research groups come up with different solutions to solve and assist with automated breed classification with the use of machine learning and technology [3].

Smith's group [4] researched that standard Convolutional Neural Networks (CNNs) can identify breeds but are typically "data-hungry," requiring thousands of training images and high-end GPUs. A lightweight system was needed to assist organizations operating on standard consumer hardware. One of these research groups is Koch's group [5], which invented a foundational approach using Siamese neural networks to help with One-Shot Learning and Metric Learning. The network consists of a twin architecture to learn meaningful feature representations from highly limited data, effectively identifying new classes from a single reference image rather than relying on massive datasets [6]. The model is highly efficient and suitable for environments with strict hardware limits.

Several research groups have developed smart systems to assist with similarity-based learning by enhancing metric-based feature extraction. Chicco's group [7] proposed a comprehensive overview of Siamese neural networks that uses distance-based metrics for image-matching and biometric tasks. Their experiments showed effective adaptation across various image-matching tasks, operating well in environments where data availability is strictly limited, although challenges remain with standard consumer-grade hardware optimization [8].

Image classification technologies have achieved progress but face ongoing difficulties because specific gaps in their systems remain unaddressed. The system requires complete retraining to work with new dog breeds because existing systems need extensive training and they require new datasets that contain all information about dog breeds [9]. The standard CNN systems need improved performance for their small-scale applications while Metric Learning methods require further development to enable their use in real-world dog classification tasks. The existing methods need to be improved through research development because they create solutions that depend entirely on advanced technological equipment and powerful GPUs which are used in data centers.

The dataset consists of a manually collected "micro-dataset" that contains 5 to 10 images for each dog breed which the researchers collected through smartphone photos and public domain sources to create an extreme data-scarcity simulation environment [10]. The system will receive input data that will be processed into positive and negative image pairs. The process will use a Lightweight Siamese-CNN architecture that employs MobileNetV2 as its feature extraction backbone to extract detailed information about canine features from the image. The model uses Euclidean distance measurements to determine how similar features are which it uses to classify input data based on whether the images show the same or different dog breeds.

III. METHODOLOGY

The methodology of this study is structured into four primary phases: Data Preprocessing, Network Architecture Design, Model Training, and Evaluation. The objective is to develop a lightweight Siamese Neural Network capable of One-Shot Learning for dog breed classification.

A. Data Collection Techniques

The data collection strategy for "Breederify" was deliberately designed to simulate the extreme data-scarce environments typical of local animal shelters and veterinary clinics. Rather than relying on massive, pre-packaged academic datasets, this study utilized a purposive manual collection technique to construct a custom "micro-dataset

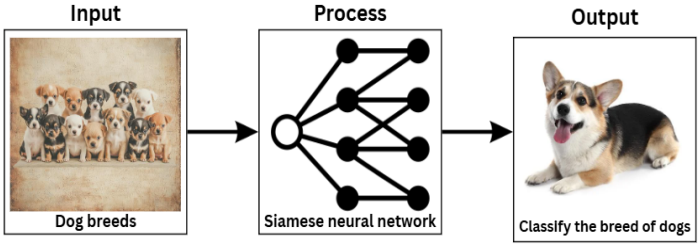


Figure 1: Conceptual Framework

The dataset consists of a manually collected micro-dataset of images which show multiple dog breeds through different camera angles and various lighting conditions. The system will utilize input data which includes both positive and negative image pairs together with actual new images that will be processed for live comparison. The Siamese Convolutional Neural Network will use a pre-trained MobileNetV2 backbone to process the image pairs. The twin network architecture extracts fine-grained canine features to generate normalized feature embeddings. The model output uses Euclidean Distance measurement to compare extracted features which the system uses to classify input data and determine whether the specific dogs under review are present..

B. Data Preprocessing

The study preprocesses the manually collected dataset by programmatically converting all native smartphone photographs from .HEIC into standardized .JPG formats to ensure compatibility with standard computer vision libraries. Because libraries such as OpenCV inherently read images in a Blue-Green-Red (BGR) format, a color correction pipeline is applied to convert all images into the standard Red-Green-Blue (RGB) color space required by the MobileNetV2 backbone. To manage the hardware constraint, all images are uniformly resized to a resolution of 224x224 pixels and have their pixel values normalized to a range between 0 and 1. Unlike standard Convolutional Neural Networks, this dataset is programmatically restructured into positive and negative image pairs to facilitate similarity-based learning. To prevent overfitting on the constrained "micro-dataset," robust data augmentation including random horizontal flipping, rotations, and zooming is integrated directly into the pipeline. These preprocessed image pairs are then stored in a list and converted into NumPy arrays, resulting in uniform image dimensions and consistent data scales ready for the Siamese Neural Network to utilize.

C. Model Classification

The model classification for Breederify utilizes a Metric Learning approach through a Siamese Neural Network to differentiate between dog breeds. Instead of assigning fixed categorical labels, the system is designed for One-Shot Learning, enabling it to identify a breed from just a single reference image by calculating the Euclidean Distance between feature embeddings. By employing a MobileNetV2 backbone for feature extraction and a calibrated similarity threshold, the architecture successfully classifies inputs as either a "MATCH" or "DIFFERENT" breed. This classification method achieved a True Accuracy of 92.50% and an Equal Error Rate (EER) of 5.00%, validating its reliability in data-scarce environments.

IV. EXPERIMENTAL RESULTS

Table 1. SUMMARY OF THE MODEL

Layer (Type)	Output Shape	Param #	Connected to
left_image (InputLayer)	(None, 224, 224, 3)	0	-
right_image (InputLayer)	(None, 224, 224, 3)	0	-
functional(Functional)	(None,128)	34,436,800	left_image[0][0], right_image[0][0]
distance_layer(Lambda)	(None,1)	0	functional[0][0],functional[1][0]

Total params: 101,206,594 (386.07 MB)
Trainable params: 33,384,896 (127.35 MB)
Non-trainable params: 1,051,904 (4.01 B)
Optimizer params: 66,769,794 (254.71 MB)

Table 1 shows the Breederify project utilizes a lightweight siamese convolutional network architecture with a MobileNetV2 backbone to achieve one-shot dog breed identification in extreme data-scarce environments. The model is optimized for standard consumer-grade hardware, featuring a total of 101,206,594 parameters (386.07 MB), with 33,384,896 parameters (127.35 MB) being trainable to capture fine-grained canine characteristics. By processing image pairs through a twin-network structure and utilizing L2 Normalization to stabilize feature embeddings, the system evaluates similarity via Euclidean Distance. Ultimately, the architecture achieves a True Accuracy of 92.50%, providing animal welfare organizations with a viable, low-cost automated tool for rapid breed cataloging.

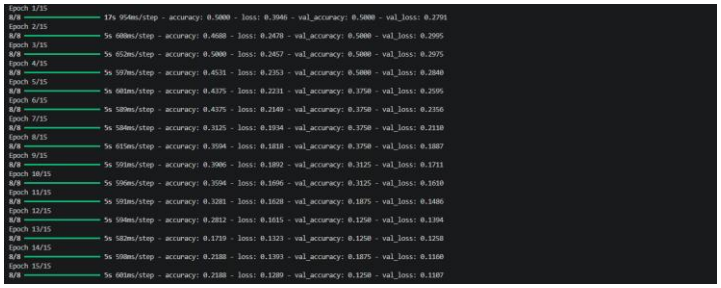


Figure 2. Training Log of the Model

Figure 2 illustrates the training trajectory of the Siamese Neural Network over 15 epochs, beginning with a 50.00% balanced accuracy and 0.3946 loss. As training progressed, validation loss reached a stable low of 0.1107, indicating the network successfully optimized feature embeddings to differentiate between breeds. Stability was maintained by implementing L2 Normalization to mitigate "exploding loss" and using data augmentation to minimize overfitting within the limited "micro-dataset". By the final epoch, the model established a consistent performance level for reliable similarity-based classification.



Figure 3. Evaluation Accuracy of the Model

The Breederify project develops a lightweight system which implements a siamese convolutional network that uses MobileNetV2 as its base model to enable one-shot dog breed identification under conditions of restricted training data. The methodology uses uniform 224x224 image resizing and L2 Normalization and strong data augmentation methods to meet stringent hardware requirements and limited available data. The experimental results confirmed the method's effectiveness which achieved a True Accuracy of 92.50% and a 5.00% Equal Error Rate when tested with a calibrated similarity threshold.

V. CONCLUSIONS AND FUTURE WORKS

The study shows that researchers can identify dog breeds with high accuracy using a one-shot method in extremely challenging conditions by applying their lightweight siamese convolutional network system which uses MobileNetV2 as its main technology. The research shows that an affordable automated system which enables animal welfare organizations to operate without specialized equipment achieved a True Accuracy of 92.50% and an EER of 5.00% based on its tests with a "micro-dataset" that contained 5–10 images for each breed. Future works will focus on expanding the dataset to 30–40 images per breed and implementing dynamic thresholding to further improve accuracy and reduce False Positives that occur between breeds with similar characteristics.

model's stability across various scenarios, such as different weather conditions and times of day.

Future improvements for Breederify could involve expanding the current "micro-dataset" to include more diverse dog breed images that show different breed characteristics through different angle views and different lighting conditions and different types of background environments. The model will achieve better results through preprocessing enhancements that will use dynamic similarity thresholding to decrease false positive rates during breed matching. The system will receive real-world deployment testing through its lightweight framework which operates on mobile devices and edge computing systems. The one-shot learning method will improve through additional research which will create a dependable portable tool that field workers can use in animal shelters and veterinary clinics.

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